



Short communication

Microstructure and mechanical properties of in-situ MC-carbide particulates-reinforced refractory high-entropy $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ matrix alloy composite

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ABSTRACT

Refractory high-entropy $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ alloy matrix composite with MC-carbide particulates-reinforced was prepared by arc melting. It is found that $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTiC}_{0.1}$ and $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTiC}_{0.3}$ alloys consist of one disordered body-centered cubic (BCC) solid solution phase as the matrix phase and MC carbide phase. MC carbide is enriched with Zr and Hf due to the higher binding strength and without any Mo. Noticeable strengthening from the carbide is not observed for $\text{C}_{0.1}$ and $\text{C}_{0.3}$ alloys while the compressive plasticity is improved slightly due to the decrease of solution strengthening for the matrix BCC disordered solid solution phase.

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1. Introduction

Metal matrix composites containing carbide, nitride, boride as a rigid ceramic reinforcement are a unique class of engineering materials which combine the ductility and toughness of matrix alloys with the high strength and high modulus of the ceramic reinforcement [1,2]. In all ceramic reinforcement, refractory carbide such as ZrC, HfC, NbC and TaC have recently drawn considerable attention in research since they show unique combination of properties such as high-melting temperature, high hardness, good high-temperature strength and good conductivity [3,4]. Thus refractory carbides are usually as high-temperature structural materials in the form of hard constituents in metal matrix composites.

In recent years, considerable research has been performed on high entropy alloys (HEAs) [5–7]. It was reported that HEAs have simple phase constitutes and exhibit excellent properties [8–10]. In all reported HEAs, refractory HEAs that are exclusively based on refractory metals have received great interest and been widely studied with the aim of increasing the mechanical properties at

high temperature. They usually form a single disordered solid solution phase or the disordered solid solution phase is the dominant phase [10–17]. In this paper, the previous reported refractory HEAs of $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ is selected as the master alloy, which forms a single BCC phase and possesses a certain compressive plasticity (the compressive fracture strain is about 24.61%) [18]. Carbon elements are added into this master alloy to prepare the refractory high-entropy alloy $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ matrix composites. The in situ MC-carbide ceramic particulates-reinforced refractory high-entropy alloy $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ matrix composites are firstly prepared successfully.

2. Experimental procedures

The samples of $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTiC}_x$ ($x = 0.1$ and 0.3 , denoted as $\text{C}_{0.1}$ and $\text{C}_{0.3}$) were prepared by arc melting the pure metals and carbon power under pure argon atmosphere in a water-cooled copper crucible. The carbon power was clad using the aluminum foil to avoid being blown away. All samples were melted five times separately to ensure the chemical homogeneity. The crystal structure was investigated by X-ray diffraction (XRD) in a range of $2\theta = 20\text{--}100^\circ$. The microstructure was studied using scanning electron microscopy (SEM) and transmission electron

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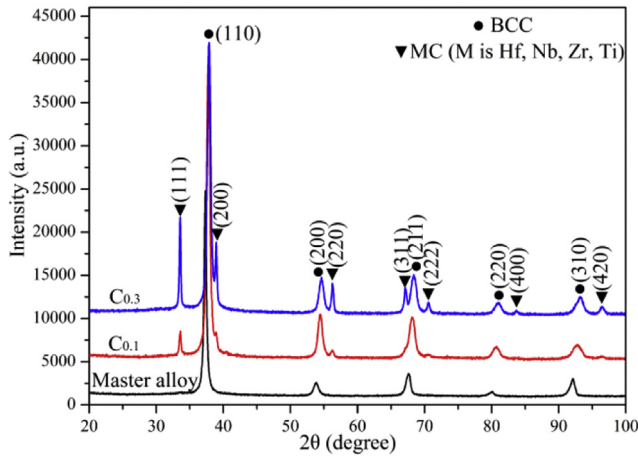


Fig. 1. XRD patterns of $C_{0.1}$ and $C_{0.3}$ alloy.

microscopy (TEM). The chemical compositions were analyzed by the SEM equipped with energy dispersive spectrometry (EDS). The mechanical properties were measured under compressive loading in an Instron 5569 testing machine at a constant strain rate of $1 \times 10^{-3} \text{ s}^{-1}$ at room temperature. The test specimens are in the shape of a cylinder ($\phi 4 \text{ mm} \times 6 \text{ mm}$).

3. Results and discussion

The XRD patterns of the master alloy, $C_{0.1}$ and $C_{0.3}$ alloys are shown in Fig. 1. As reported in other paper, the master alloy $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ was composed of one disordered body-centered cubic (BCC) solid solution phases. $C_{0.1}$ and $C_{0.3}$ alloys consist of predominant BCC phase (space group Im-3m (229)) which match well with ZrTiNb (#65-7192) and some carbide which corresponds nicely to HfC (#65-8751, space group Fm-3m (225)). The diffraction intensity of carbide increases with more addition of carbon which indicates more carbide forms. The lattice parameters of BCC phase in $C_{0.1}$ and $C_{0.3}$ are 337.05 pm and 335.92 pm, respectively and the lattice parameter of BCC phase in the master alloy $\text{Mo}_{0.5}\text{NbHf}_{0.5}\text{ZrTi}$ is 340.63 pm which indicates that the addition of carbon decreases the lattice parameter of BCC phase. This is attributed to carbon atoms occupy some metal atoms to form carbide and carbon atoms have different binding energy with five metals elements which causes the decrease of the configurational randomness of BCC phase (this will be verified by the following chemical compositions analysis) and thus decreases the degree of the lattice distortion.

Microstructure of $C_{0.1}$ and $C_{0.3}$ alloys is shown in Fig. 2. The chemical compositions of Mo, Nb, Hf, Zr and Ti in different regions are listed in Table 1 since the measured value of carbon is not exact using EDS due to its light atomic weight. $C_{0.1}$ and $C_{0.3}$ alloys exhibit dendritic structure and small amount of white precipitates distribute in the dendrites. However, only two phases of carbide and BCC are detected from the XRD patterns. The chemical

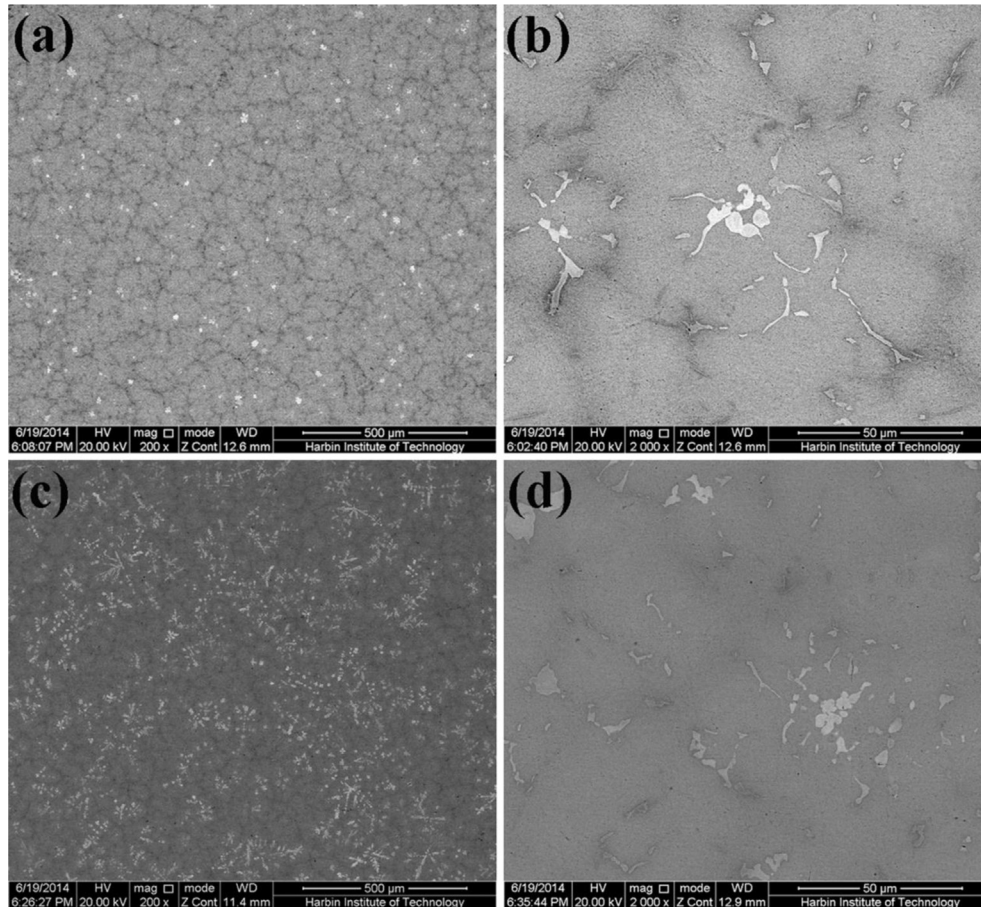


Fig. 2. SEM-BSE images of $C_{0.1}$ and $C_{0.3}$ alloy: (a) and (b) $C_{0.1}$; (c) and (d) $C_{0.3}$.

Table 1
The chemical compositions (at.%) of $C_{0.1}$ and $C_{0.3}$ alloy.

Sample	Region	Mo	Nb	Hf	Zr	Ti
$C_{0.1}$	Average	11.26	24.77	14.82	24.20	24.94
	Dendrites	11.89	26.16	14.65	22.49	24.79
	Interdendrites	11.12	20.14	12.73	28.51	27.50
	White precipitates	0	5.68	39.26	46.93	8.16
$C_{0.3}$	Average	12.21	24.80	14.29	23.87	24.83
	Dendrites	14.09	32.10	10.80	17.48	25.47
	Interdendrites	8.44	15.32	17.47	39.02	19.78
	White precipitates	0	8.68	33.23	48.07	10.02

compositions shows that the white precipitates are enriched with Zr and Hf and the concentration of Mo is zero. Mo element exists in the dendrites and interdendrites. The master alloy $Mo_{0.5}NbHf_{0.5}ZrTi$ formed a single BCC phase which indicates that Mo elements can displace mutually with other four elements. It can be deduced that the white precipitates are carbide phase and dendrites and interdendrites are BCC phase. The mixing enthalpy between carbon and Mo, Nb, Hf, Zr and Ti is -67 , -102 , -123 , -131 and -109 kJ/mol, which indicates that carbon has the strongest binding strength with Zr and Hf and tend to form (Zr, Hf)C

preferentially. Mo has the weakest binding strength with C and MoC has different crystallographic structure. According to the binary alloy phase diagrams, five metals can form MoC, NbC, HfC, ZrC and TiC carbide with carbon element while NbC, HfC, ZrC and TiC belong to the same cubic crystallographic structure with space group Fm-3m (225) with the lattice parameters of 4.471 Å, 4.620 Å, 4.639 Å and 4.670 Å, while the MoC has a hexagonal crystallographic structure with space group P-6m2 (187). Thus the white precipitates are MC-typed carbide (M is Hf, Nb, Zr and Ti).

The microstructure and crystallographic structure are also analyzed by TEM as shown in Fig. 3. The precipitate (region B) distributes in the matrix (region A). The corresponding selected area electron diffraction patterns of A and B show that the matrix is a disordered BCC phase and the precipitate has a face-centered cubic (FCC) crystallographic structure. This result is consistent with the previous analysis. It can be concluded that $C_{0.1}$ and $C_{0.3}$ alloys are composed of one disordered BCC phase and MC-typed carbide with FCC crystallographic structure enriched with Zr and Hf and containing small amount of Nb and Ti.

Fig. 4 shows the compressive stress–strain curves of master alloy, $C_{0.1}$ and $C_{0.3}$ alloys. The yield strength, peak strength and fracture strain of $C_{0.1}$ and $C_{0.3}$ alloys are (1183 MPa, 2139 MPa,

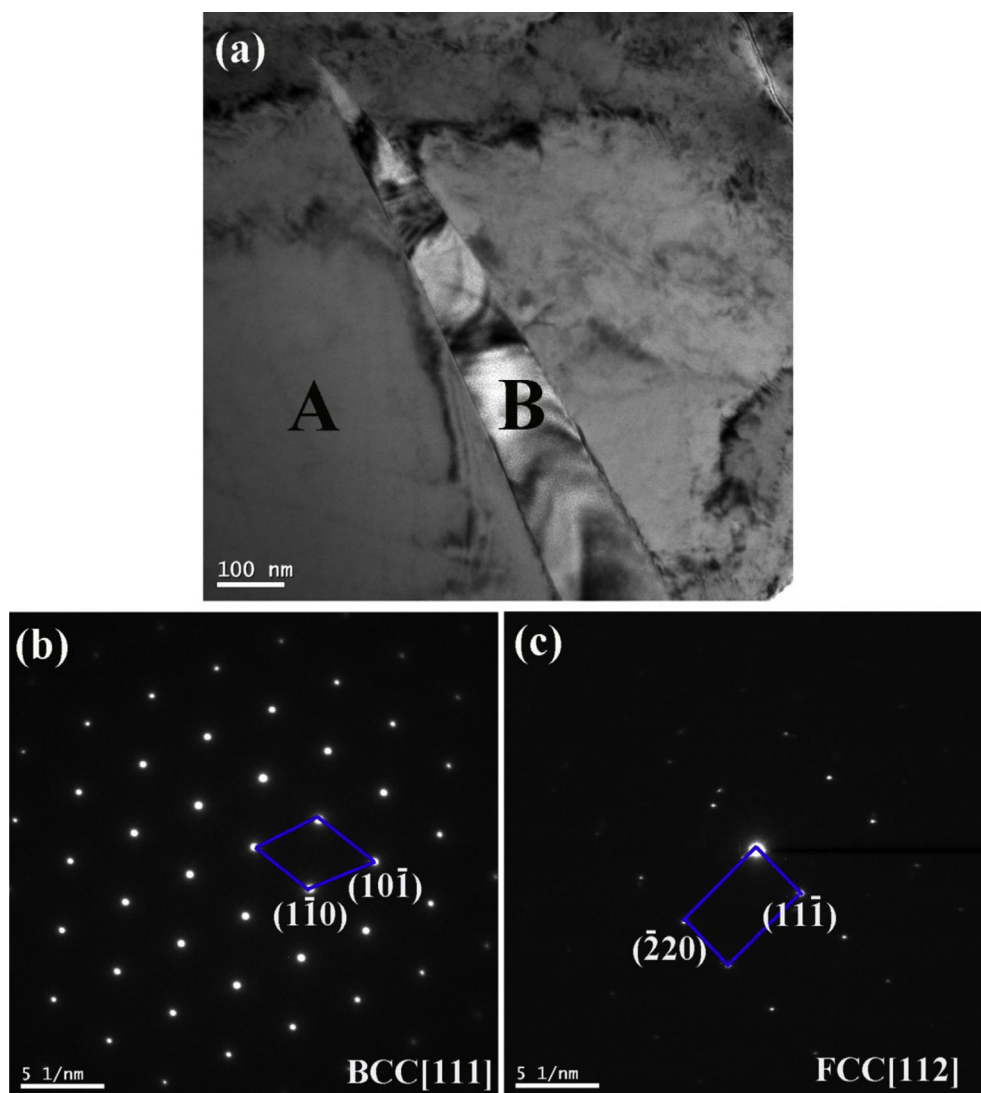


Fig. 3. (a) TEM bright-field image of $C_{0.1}$ alloy, (b) and (c) are the corresponding selected area diffraction patterns of region A and B in (a).

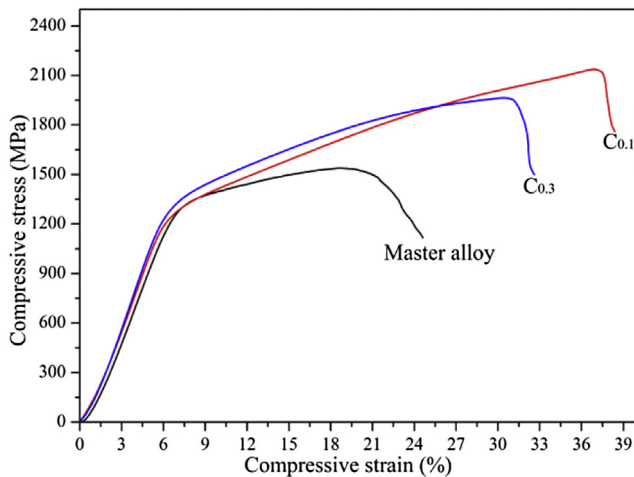


Fig. 4. Compressive stress–strain curves of master alloy, $C_{0.1}$ and $C_{0.3}$ alloy.

38.39%) and (1201 MPa, 1965 MPa and 32.62%). Compared with the parent alloy $Mo_{0.5}NbHf_{0.5}ZrTi$ (the yield strength, peak strength and fracture strain are 1176 MPa, 1538 MPa and 24.61%, respectively), the addition of carbon has no obvious effect on the yield strength but increases the peak strength and work hardening rate. And the plasticity is also improved slightly. The following possible reasons for the little change of the yield strength and the improvement of plasticity: the strengthening mechanism of the parent with single BCC phase is solution strengthening while the strengthening mechanism of $C_{0.1}$ and $C_{0.3}$ alloys with BCC phase and carbide phase is solution strengthening and second-phase strengthening. The addition of carbon leads to the decrease of solution strengthening effect and produces second-phase strengthening. Carbon atoms occupy some metals atoms from the BCC phase to form carbide, which will decrease the degree of chemical compositions for the BCC solid solution phase. Thus the solution strengthening effect is reduced and the plasticity of the matrix BCC phase is improved. Work hardening arises from the interaction of dislocations with amongst themselves or with other defects such as second-phase particles. The higher work hardening rate of $C_{0.1}$ and $C_{0.3}$ alloys is mainly attributed to the hardening from the carbide which will produce a turbulent and complex deformation pattern around them leading to the increase of the dislocation density.

4. Conclusions

In summary, carbon element with molar ratio of 0.1 and 0.3 is added into refractory high entropy alloy $Mo_{0.5}NbHf_{0.5}ZrTi$ for preparing the in-situ MC-carbide ceramic particulates-reinforced $Mo_{0.5}NbHf_{0.5}ZrTi$ matrix composite by arc melting successfully. $Mo_{0.5}NbHf_{0.5}ZrTiC_{0.1}$ and $Mo_{0.5}NbHf_{0.5}ZrTiC_{0.3}$ alloys are composed of BCC solid solution phase and particulates-reinforced phase MC-

carbide. MC carbide with FCC crystal structure is enriched with Zr and Hf and containing a small amount of Nb and Ti. Both the compressive strength and plasticity of $Mo_{0.5}NbHf_{0.5}ZrTiC_{0.1}$ and $Mo_{0.5}NbHf_{0.5}ZrTiC_{0.3}$ alloys are improved compared with the master alloy $Mo_{0.5}NbHf_{0.5}ZrTi$.

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